IWACU RWANDA

What the project lacks to become a real, nationwide system

Gap analysis of the uploaded project, with the national geography the system must model and the steps to build it

Subject	Archive iwacu-rwanda-28-oct-2026-10-31-52 (57 files)
Date	29 October 2026
Method	Every source file, SQL file, script, document and the three PDFs in public/assets/aistudio were read. The code was read, **not run**. National boundary facts come from public pages (section 11), which I read but did not download data from.
Scope	Technical and data gaps only. Findings are stated as they are in the files.
Severity	**Critical**: untrue or unsafe as it stands. **High**: blocks national use. **Medium**: needed for a quality product.

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1. Bottom line

> The archive is a working interface and a location-matching idea. It is not yet a real system. It contains **no real geography**: the only boundary in the whole project is one placeholder rectangle named “Sample Village”. Every answer a citizen can get today is either computed against that rectangle or hard-coded.

What a citizen sees today is made from: one fixed result (“Sample Village”, 85.5 %), eight invented people, and hand-typed lists of 35 sectors, 11 cells and 7 villages. The admin area does not check a password, and several write endpoints can be called by anyone.

To go from a one-district prototype to a system that covers the whole country, six things are missing:

- **The national boundary dataset**: national outline, 5 provinces, 30 districts, 416 sectors, the cells and the villages, loaded, checked and versioned (sections 5 and 6).
- **One official code scheme and real parent links**, so the chain village → cell → sector → district → province is read from data and not guessed from text (findings B1–B5).
- **A locate engine tested on national data that never invents an answer**; where data is not loaded yet it says so (section 7).
- **Real login, scoped permissions and closed endpoints** (findings E1–E8).
- **A way for true data to arrive after launch**: coverage tracking, verification, citizen corrections and new boundary versions (section 8).
- **Tests, monitoring, rate limiting and backups** (finding F1).

Scoreboard

Area	Today (from the files)	Needed for a national system
Boundaries	1 placeholder rectangle	National outline, 5 provinces, 30 districts, 416 sectors, about 2,150 cells, about 14,800 villages
Hierarchy and codes	3 incompatible code schemes; hierarchy guessed from text	1 official scheme; parent links; province included
Place lists	5 provinces, 30 districts complete; 35 sectors, 11 cells, 7 villages typed by hand; the rest generated as fake entries	Lists generated from loaded boundary data only
Locate engine	Written, never tested on real polygons; returns a fixed fake result on any failure	Tested at national size; honest “not available yet”
Access control	Login checks nothing; open write endpoints; private fields readable by anyone	Server-checked sessions and scopes; audit
Offline	Boundary cache never filled	Regional packs with a tested cache
People and contact data	8 invented people in code	Loaded later through verification (separate layer)
Tests and operations	None in the archive	Tests, CI, health, rate limits, monitoring, backups

2. What the archive contains

Part	What is there	State
Public page (src/app/page.tsx)	One-button location, consent dialog, result card, AI search box. Kinyarwanda copy.	Interface complete. Shows “Kigali”, “Gasabo”, “Kacyiru” as defaults when data is missing.
/api/locate	GPS → PostGIS function → hierarchy → leaders.	Logic written. Returns a fixed fake result on any failure. Never run on real polygons.
/api/leaders, /api/report-number, /api/ai-router	Read, edit, create, delete people; citizen reports; AI search.	Create, delete and reports use server memory only. No authentication.
Admin pages	Login and dashboard.	Interface only. No real authentication.
sql/schema.sql	4 tables, 1 view, 4 functions, row-level security.	No boundary versions, sources, verification or names table.
scripts/ingest-boundaries.ts	Loads GeoJSON one feature at a time.	Not usable for a national load (A5).
public/offline/…geojson	One rectangle, “Sample Village”.	Placeholder.
public/sw.js, manifest.json	Service worker, PWA manifest.	Boundary cache never filled; icon files missing.
src/lib/rwanda-geo.ts	5 provinces, 30 districts, 35 sectors, 11 cells, 7 villages, 10 emergency numbers.	Typed by hand.
src/lib/mock-data.ts	8 invented people, 1 sample report.	Fake data.
Documents	README, DEPLOYMENT, SECURITY_AUDIT, FOLDER_STRUCTURE; three PDFs (project specification, master specification and delivery plan, Kinyarwanda copy bank).	Several statements do not match the code (F4).

Archive versus specification

The master specification (27 Oct 2026) and the project specification (28 Oct 2026) describe a larger system. These items appear in the specifications and are **not in the archive**: /api/health, /api/emergency, /api/admin/*, /auth/callback, src/middleware.ts, sql/migrations, apply_all.sql, seed files, tests, CI, Redis rate limiting, the point_in_rwanda function, the tables data_sources, boundary_history, contacts and emergency_contacts. The master specification itself marks everything “unverified” and Stage 4 (national boundaries) as “not started”. This report judges what is in the archive.

3. How Rwanda is divided: the geography the system must model

3.1 The five levels

Level	Kinyarwanda	Count, 2006/2012 frame	Count, 2022 census frame	Note
Province	Intara	5	5	Kigali City, Northern, Southern, Eastern, Western
District	Akarere	30	30	Kigali 3, Southern 8, Western 7, Northern 5, Eastern 7
Sector	Umurenge	416	416	
Cell	Akagari	2,148	2,146	Counts differ between frames
Village	Umudugudu	14,837	14,774	Smallest administrative unit; 63 fewer in the newer frame

The 2,148 / 14,837 figures appear in national statistics publications and in the descriptions of the national GIS layers. The 2,146 / 14,774 figures come from a summary of the 2022 census metadata. **Two published counts mean the boundaries changed or were re-mapped between the two frames.** The project hard-codes the older figures (in the AI prompt) and has no way to hold two versions. I did not confirm which frame is treated as current.

3.2 A check against the project's own lists (Kigali City)

District	Sectors (public sources)	Cells	Villages	Sectors in rwanda-geo.ts
Gasabo	15	not checked	not checked	15 (complete)
Kicukiro	10	41	333	5 (missing Gatenga, Gikondo, Kigarama, Masaka, Nyarugunga)
Nyarugenge	10	47	350	5 (missing Kanyinya, Kimisagara, Mageragere, Nyakabanda, Rwezamenyo)

Nyarugenge figures: district website. Kicukiro figures: public encyclopedia entry. Hand-typed lists are already incomplete in the capital city; at national size they cannot be maintained by hand.

3.3 Properties of the real geography that the design must handle

- **Names are not unique.** "Gasabo" is a district, a cell (Rutunga sector) and the name of several villages in other districts. "Ngoma" is a district in the Eastern Province and a sector in Huye (the project lists both). Units must be identified by ID and shown with their full path.
- **Village size varies a lot.** NISR survey samples record villages from about 50 to about 2,000 households. A dense Kigali village and a large rural village behave very differently under one GPS accuracy rule.
- **Four neighbouring countries.** Rwanda borders Uganda, Tanzania, Burundi and the DR Congo. A rectangle around Rwanda contains parts of all of them, for example Goma and Kabale.
- **Lakes.** Whether source polygons include lake surface must be checked. The project itself lists a lake emergency number (110), so users on or near lakes are expected.
- **Change over time.** Rwanda went from 12 provinces to 5 in 2006 and from 106 districts to 30. Cell and village counts differ between the 2012 and 2022 frames. New settlements appear. The system needs versions, not one fixed set.
- **Hierarchy is strict.** Each village belongs to exactly one cell, each cell to one sector, and so on up to the province. Data and code should enforce this, not assume it.

4. Findings: what is missing

Each finding gives what the files show and what is needed. IDs are used in the other sections. Statements about code behaviour come from reading the code.

A. Geographic data (the foundation)

A1
CRITICAL

No real boundary data exists in the project

In the files. The only boundary in the archive is public/offline/gasabo-boundaries.geojson: one rectangle (longitude 30.12–30.14, latitude −1.95 to −1.93) named "Sample Village", code RW-01-01-01-001, collection name "Gasabo_ADM5_MVP_Placeholder". SECURITY_AUDIT.md lists replacing it as open work. schema.sql creates empty tables; nothing is loaded until someone runs the ingest script with a real file.

Needed. The complete national set: national outline, 5 provinces, 30 districts, 416 sectors, the cell layer and the village layer (sections 5 and 6), loaded, versioned and checked. Until this exists, nothing else in the project is a result.

A2 CRITICAL

Nothing checks that the boundaries are consistent

In the files. The app defines error states for gaps and overlaps (POINT_IN_GAP, POINT_IN_OVERLAP), but no code or SQL checks the dataset itself: not that children sit inside parents, that villages do not overlap, that the villages of a cell add up to the cell, or that counts match official counts. The master specification names a boundary_integrity_report() function; it is not in the archive.

Needed. An integrity report that runs after every load and blocks publication on errors (section 6, step 5).

A3 HIGH

No record of where the boundaries came from

In the files. The boundaries table has no source, publisher, version, retrieval date, licence or verification-status column. The master specification requires data_sources, boundary_history and verification_status; none exists in schema.sql.

Needed. A data_sources table, a version per load and a verification status per unit. Public queries return verified units only.

A4 HIGH

Boundaries cannot be versioned

In the files. The table has valid_from / valid_to, but scope_code is UNIQUE and leaders.scope_code references it, so an old and a new version of the same unit cannot exist together. The ingest function also overwrites the geometry in place on conflict (version + 1), so the old shape is lost.

Needed. Separate a permanent unit identity (never reused) from its geometry versions. Keep old versions. Keep a mapping table for splits and merges. This matters because published counts already differ between sources (section 3.1).

A5 HIGH

The ingest script cannot do a national load

- **Level names.** The lookup knows village, cell, sector and district only. Any other value, including “province”, “ADM1” or “country”, silently becomes “village”. A province polygon would be stored as a village, and the locate function would then match it for every point in that province.
- **No reprojection.** The header says files “will be transformed”; the code does not do it. The public source layers are in the national Transverse Mercator projection (REMA service) or Web Mercator (NISR sector service), not WGS84.
- **parent_id is never set.** The hierarchy exists only in the text of the code.
- **One call per feature**, with the geometry sent as text. No dry run, no transaction per level, no resume after a failure, no summary of what changed.
- **Codes must already be in the file.** Every feature needs scope_code. Public layers carry their own IDs and names, so a mapping step is needed.
- **Fallback branch** (direct upsert) is marked in a code comment as one that “will fail”.

Needed. A rewritten pipeline (section 6): staging tables, reprojection, validation, code mapping, parent linking, bulk load per level, and a report.

A6 HIGH

Only villages are matched; the other four levels have no working geometry path

In the files. calculate_location_confidence() filters to level = 'village'. Cell, sector, district and province polygons are used for nothing, and nothing verifies that a village’s parents are right. The offline engine looks for a cell polygon in the cache that the SQL path does not use.

Needed. All five levels loaded with geometry. Parent levels used for validation, for map display and for the answer “you are in a district that is not loaded yet”.

A7 MEDIUM

No national outline; “outside Rwanda” is a rectangle

In the files. OUTSIDE_RWANDA is decided by a box (longitude 28.85–30.90, latitude −2.85 to −1.05). Goma (DR Congo) and Kabale (Uganda) are inside that box, so a person there is not told they are outside Rwanda; they get a “gap” answer or, in the fallback, “Sample Village”. The master specification names a point_in_rwanda function; it is not in the archive.

Needed. A real national outline polygon and an exact point-in-polygon test. Keep the box only as a cheap first filter.

A8 MEDIUM

No place-name table or search

In the files. Names exist only as name_kiny / name_eng on each boundary: no alternative spellings, no search index, no way to separate duplicate names (section 3.3). The only search in the archive is substring matching over the hand-typed lists; its answers hard-code “Gasabo”.

Needed. A names table (unit ID, name, language, variant), a search index, and results shown with the full path (village, cell, sector, district) so users can pick the right one.

A9 MEDIUM

No map and no display geometry

In the files. README lists MapLibre GL; it is not in package.json and no page draws a map. There are no simplified geometries or tiles, so full-resolution polygons would be the only option for display or offline use.

Needed. A simplified geometry per level, or vector tiles, for map display and offline packs.

A10 MEDIUM

Place lists are typed by hand, incomplete, and padded with synthetic entries

In the files. rwanda-geo.ts holds 5 provinces and 30 districts (complete), but only 35 sectors, 11 cells and 7 villages. Gasabo has all 15 sectors; Kicukiro 5 of 10; Nyarugenge 5 of 10; four more districts have 2–3 sample sectors; the other 23 districts have none. For any missing unit, getSectorsByDistrict / getCellsBySector / getVillagesByCell generate made-up entries such as “Umurenge w’Icyitegererezo 1”, so screens always look filled. The admin forms use these lists, so an admin can attach data to a unit that does not exist.

Needed. Lists generated from loaded boundary data, never typed. Screens show real units only and an honest “not available yet” for the rest.

B. Hierarchy and codes

B1 CRITICAL

Three incompatible code schemes in one code base

In the files. The same code text means different things in different files:

Where	Example	Meaning in that place
schema.sql, sample polygon, DEPLOYMENT.md	RW-01-01-01-001	Village code with 5 parts (no cell level)
rwanda-geo.ts, mock data, admin forms	RW-01, RW-01-01, RW-01-01-01, RW-01-01-01-01, RW-01-01-01-01-01	Province, district, sector, cell, village (6 parts for a village). RW-01 is Kigali City.
/api/locate (hierarchy and fallback), .env.example, DEPLOYMENT.md	RW-01 (2 parts), 3 parts, 4 parts	District, sector, cell. RW-01 is “Gasabo district”.

Needed. One scheme, defined once (section 6, step 2) and used by the database, API, admin forms and import.

B2 HIGH

Codes are invented, not official

In the files. Codes follow the order in which units were typed (for example Kacyiru = 01, Kimironko = 02). The national statistics office publishes its own numeric IDs: 1 digit for the province, 2 digits for the district (province digit first), 4 digits for the sector (province, district, then two digits). Invented codes would have to be replaced later, and every record that points to them re-linked.

Needed. Adopt the official IDs as the public code of each unit. Extend the same pattern to cells and villages once the official cell and village code tables are obtained. Keep a permanent internal ID as well, so codes can change without breaking links.

B3 HIGH

Province is missing from the answer; defaults are hard-coded

In the files. The API response has village, cell, sector and district only. The home page prints “Umujyi wa Kigali” as the province for everyone and falls back to “Gasabo” and “Kacyiru” when a field is missing. The confidence badge defaults to “Twabyizeye cyane” (very sure) when there is no data.

Needed. Province in the response. No default text for any geographic field; an unknown level is shown as unknown.

B4 MEDIUM

Parent links are never filled

In the files. boundaries.parent_id exists but no code sets it. The ltree path is made by replacing “-” with “.” in the code, so if the code scheme changes, paths and hierarchy break silently.

Needed. Parent links set and checked at load time; hierarchy read from them.

B5 MEDIUM

Scope checks use text prefixes

In the files. Admin scope and the leaders query use LIKE 'prefix%' on text codes although an ltree column exists. Text prefixes are fragile when codes are not fixed-width, and the query parameter is passed to LIKE unescaped (% and _ act as wildcards).

Needed. Use the ltree ancestor test or parent IDs for scope; escape input.

C. Locate engine

C1 CRITICAL

The locate API returns a fixed fake answer when anything fails

In the files. In /api/locate, if Supabase is not configured, or the database call returns any error, or the offline calculation returns an error (for example any point outside the placeholder rectangle), the route returns status “success” with Sample Village, 85.5 % confidence, Kacyiru, “Akagari k’Urugero” and the mock leaders. In the default setup, every location inside Rwanda’s bounding box with accuracy up to 150 m gets this answer.

Needed. No fallback on public paths. A database or data problem returns DATABASE_ERROR or DATA_NOT_AVAILABLE. Both texts exist in errors.ts, but the locate route never returns DATA_NOT_AVAILABLE.

C2 HIGH

The hierarchy is guessed from the text of the code

In the files. buildHierarchy() splits the village code on “-” and takes the first 4, 3 and 2 parts as cell, sector and district. With the 6-part codes of rwanda-geo.ts every level is shifted one step (the “cell” is the sector, the “sector” is the district, the “district” is the province). The lookup also requires the level to match, so these levels come back empty and the page then shows its hard-coded defaults (B3). Each level inherits the village’s confidence number.

Needed. Read ancestors from parent links in one query and return all five levels with names and IDs.

C3 HIGH

Not tested or measured at national size

In the files. The SQL function intersects the accuracy circle with each candidate village using geography calculations. There is no benchmark, no sample of real polygons, no query plan, no timing, no load test. The national village layer has about 14,800 polygons with detailed outlines.

Needed. A performance test on the real national data: first pass by bounding box or distance, second pass exact; area measured in a metric projection (the national Transverse Mercator, central meridian 30°E, used by the source layers); a recorded response-time target and a load test.

C4 MEDIUM

Accuracy rules are untested thresholds

In the files. The model treats every point inside the accuracy circle as equally likely, rejects accuracy above 150 m, and rejects matches below 5 %. These values were not checked against real phone readings. Village size varies strongly (section 3.3), so one threshold behaves differently in a dense Kigali village and in a large rural one.

Needed. Field tests with real phones in mixed places (dense urban, rural hills, lake shore, border areas), then tuning; results tracked by area type to see where it fails.

C5 MEDIUM

Only GPS works; the other ways in are missing

In the files. README says GPS, IP or manual search. The code has GPS only: IP_GEO_API_KEY is unused and no page produces source “ip” or “manual”. A screenshot in public/assets shows a district picker that is not in this page’s code. If GPS is refused, inaccurate (above 150 m) or unavailable indoors, the user has no other path.

Needed. Fallbacks: step-by-step choice (province → district → sector → cell → village) from real data; place-name search; tap on a map.

C6 MEDIUM

Part of the circle outside every village is not reported

In the files. Percentages are scaled only when they add up to more than 100. When the circle partly lies in a gap or outside the country, the missing share is dropped silently (for example 70 % + 10 % is shown as if the other 20 % did not exist).

Needed. Report the unmatched share (gap or outside) explicitly.

C7 HIGH

Offline mode does not work as built, and cannot scale as designed

- The hook reads a cache named “iwacu-boundaries-v1”. The service worker names its cache “iwacu-boundaries-iwacu-v2-nocache-html” and never writes to it, and the page script deletes every cache whose name lacks “v2-nocache-html” on each load. The boundary file is never stored for offline use.
- The offline engine loops over every polygon and intersects each one, with no bounding-box filter or index. Acceptable for one rectangle; too slow for thousands of polygons.
- One national file at full detail is likely too large for thin connections (not measured).

Needed. Regional packs (per district or sector) of simplified polygons with a spatial index, downloaded on request, versioned, with a tested cache name.

D. Fake or unsourced data presented as real

D1 CRITICAL

Invented people are in the code as real answers

In the files. mock-data.ts holds eight people with invented names, +25078… phone numbers, e-mail addresses on gasabo.gov.rw and musanze.gov.rw, stock photos, appointment-letter references and “verified_by: Akarere ka Gasabo”, all flagged verified. They are returned by /api/locate, /api/leaders and /api/ai-router whenever the database is missing or errors. The master specification’s own rule is no invented people in any public deployment. Citizens would call these numbers.

Needed. Remove from public paths. Keep test data only behind a development switch and visibly marked. People data arrives later through verification (section 8); it is a separate layer from geography.

D2 HIGH

Emergency numbers are marked verified when the specification says three are not

In the files. rwanda-geo.ts marks all 10 numbers is_verified: true, and the emergency strip shows all of them. The master specification lists only 112, 111, 113, 3511, 110, 997 and 118 as verified and says 114, 116 and 3512 stay hidden until confirmed.

Needed. A verification status and a source stored per number; the strip shows confirmed numbers only.

D3 HIGH

For most features the “database” is server memory

In the files. Adding and deleting people, editing people when no database is set, and all citizen reports go to a variable in server memory. POST and DELETE /api/leaders never touch the database, even when it is configured. On hosting that runs several short-lived instances this data is lost or differs between requests. The report endpoint tells the citizen the report was sent to the district and sector offices; nothing is sent.

Needed. Real tables for every write; honest messages; a place where reports are actually reviewed.

D4 MEDIUM

The interface uses fields the database does not have

In the files. Types, forms and API use whatsapp_number, office_hours, vision_priorities, social_media, admin_dossier, is_verified, verified_by and availability_status. schema.sql has none of these columns, and the database PATCH path ignores them. With a real database these parts would show nothing or fail.

Needed. Decide the people/contact data model as its own layer and make schema and code match.

E. Access and data exposure (technical)

E1 CRITICAL

The admin login checks nothing

In the files. The passcode field is stored in state and never used. The person picks a tier (national, province, district, sector, cell) and a place from drop-downs; the choice is saved as JSON under “iwacu_admin_session” in the browser’s localStorage, and the dashboard trusts it. With nothing saved, the dashboard opens as a district admin of Gasabo. There is no middleware and no /auth/callback, and the login page has no sign-in call, although SECURITY_AUDIT.md describes magic-link login and redirects.

Needed. Real authentication with server-verified sessions; scope read from the database profile; every admin page and API checked on the server.

E2 CRITICAL

Write endpoints are open

In the files. PATCH, POST and DELETE on /api/leaders, and GET and PATCH on /api/report-number, have no authentication or scope check. In database mode PATCH uses the service-role key, which bypasses row-level security, so the database rules do not protect it either.

Needed. A server-side check of the signed-in admin and of the unit’s scope before any write; use the user’s own database session where possible.

E3 CRITICAL

Private data can be requested by anyone

In the files. GET /api/leaders?reveal=true&include_dossier=true returns raw phone numbers and the HR dossier (education, work history, appointment-letter references) with no check; SECURITY_AUDIT.md says it is admin-gated. In the no-database path every public response also carries the raw number in the whatsapp_number field.

Needed. Remove these parameters from public routes; separate admin routes that check the caller; no raw private fields in public objects.

E4 HIGH

The database exposes the phone column directly

In the files. schema.sql grants SELECT on the leaders table to the public anonymous role and adds a public read policy. The view v_active_leaders leaves phone_number out, but the table can be queried with the public key and returns every column.

Needed. Revoke table access for public roles; expose only the view or a function; test as the public role.

E5 HIGH

After the reveal tap, call and WhatsApp links use the masked text

In the files. LeaderCard builds tel: and wa.me links from phone_raw or, if that is missing, from the masked string. The public database path never sends phone_raw, so the call link becomes tel:078***456 and the WhatsApp link uses the digits left in the masked text. These do not reach the person and could reach a wrong number. (In the no-database path the WhatsApp link works only because the raw number leaks, see E3.)

Needed. Decide once how a public contact is delivered, and test the links on real phones.

E6 HIGH

The AI route sends the full people list to an outside service

In the files. /api/ai-router puts every leader record (name, role, area, masked phone, priorities, office hours) into the prompt to Gemini on every query. README and SECURITY_AUDIT.md describe Groq with no personal or leader data; groq-sdk is installed but unused. Facts such as “who leads X” or unit counts are answered by the model, not read from the database, and the keyword fallback hard-codes Gasabo in several places.

Needed. Use AI only to turn a sentence into a place or intent; answer from the database; no personal data in the prompt; the same rule in documents and code.

E7 HIGH

No rate limiting

In the files. The page handles HTTP 429, but nothing in the code returns it. /api/locate and /api/ai-router are public, and the second calls a paid service. The specification calls for Redis-based limits; they are not in the archive.

Needed. Server rate limits per client on all public routes; writes fail closed.

E8 MEDIUM

The audit log is never written

In the files. audit_logs and its policies exist and direct inserts are blocked, but no function or trigger inserts rows. check_version() exists and is never called.

Needed. Triggers or functions that record every change: who, what, before, after.

F. Operations and documentation

F1 HIGH

No tests, CI, health check, monitoring or backups

In the files. There are no test files, no test script in package.json, no CI configuration and no /api/health. The master specification says 131 unit tests exist in the author’s sandbox; none are in the archive. There is no error tracking, backup or restore plan, or rollback procedure.

Needed. Unit tests for geometry and hierarchy; database tests with real polygons; end-to-end tests; CI on every change; a health endpoint; monitoring; backups with a tested restore.

F2 MEDIUM

Setup instructions do not match the files

In the files. README says “npm run seed:leaders”; no such script exists. schema.sql cannot be run twice (types, policies and triggers are created without IF NOT EXISTS) and there is no migration system. public/icons is missing although the manifest points to it. package.json lists packages that nothing imports (groq-sdk, uuid, date-fns, class-variance-authority, clsx, tailwind-merge).

Needed. Numbered migrations that can be re-run; scripts and documents that match the files; unused packages removed.

F3. Documents versus code

Document says	Code does
README: AI is Groq function calling only, zero personal data	Gemini; full people list in the prompt
README: locate by GPS, IP or manual search	GPS only
SECURITY_AUDIT: admins sign in by magic link; dashboard redirects unauthenticated users	No sign-in call; localStorage session; passcode unchecked
SECURITY_AUDIT: raw phone only for the authenticated admin of the matching scope	No check
SECURITY_AUDIT: service worker caches the Gasabo boundary file for offline use	The boundary cache is never written
FOLDER_STRUCTURE: /api/admin/leaders and public/icons exist	Neither is in the archive
Master spec: nationwide from day one, no district special-cased	Gasabo in the env default (RW-01), offline file, fallback results, admin defaults and AI answers
Master spec: no placeholder boundary and no invented person ships	Both are in the archive

5. National boundary data: sources and decisions

5.1 Public sources found (read, not downloaded)

Source	What it offers	Note
NISR 2011–12 census mapping, described on the Rwanda Space Agency GIS portal	Sector and cell layers “since 2006, updated by the 2012 census mapping”. A field team collected boundaries “up to the lowest level, Umudugudu”, adjusted with 2008–09 orthophotos.	416 sectors, 2,148 cells
REMA GIS server, Administrative_Boundaries service	Layers: District, Sector, Cell, Village. Credit “RLMUA/RNRA 2012”; pages dated 2018. National Transverse Mercator projection.	DEPLOYMENT.md already points to REMA / RLMUA
RBC GIS service “Full_District_boundary_2022”	Described as boundaries built from the village boundaries established in the 2022 census.	Shows a 2022 village layer exists; downloadable form not confirmed
2022 census mapping (NISR; FAO metadata summary)	Whole country mapped with high-resolution satellite imagery (over 4 million buildings). Frame of 416 sectors, 2,146 cells, 14,774 villages.	Newer than the 2012 counts
Rwanda Space Agency GIS, Government_Assets service	Locations of district, sector and cell offices from the 2022 mapping (partly updated 2024).	Useful later for “nearest office”
Ministry of Health data portal (reference page)	Lists downloadable cell (admin4) and village (admin5) shapefiles attributed to NISR; provinces to sectors are 2006 vintage.	Version of the cell and village files not stated
NISR sector service, attribute description	Official ID scheme: province 1 digit, district 2 digits, sector 4 digits.	Basis for B2

5.2 What I could not confirm

- The licence or terms of use of each layer, and whether the 2022 village polygons are publicly downloadable.
- The attribute fields (codes, names) of the cell and village layers.
- Whether the cell, village and sector files come from the same vintage and agree with each other.
- Whether lake surface is inside the polygons, and how border areas are handled.
- Which frame (2012 or 2022) the government treats as current.

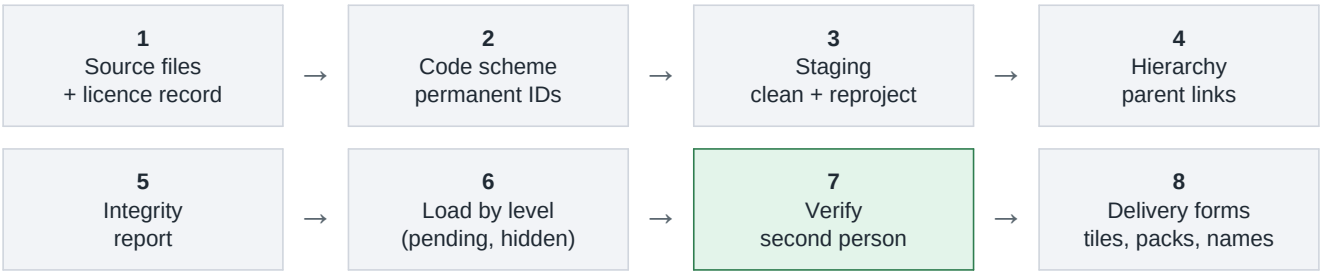
The first action is therefore to ask NISR (and the land authority if needed) for the authoritative village-level layer with its code table, the national outline, and written permission to use them.

5.3 Decisions needed from the owner

Decision	Options	Recommendation
Base vintage	2012 frame (14,837 villages) or 2022 frame (14,774)	The newest official village layer as the base; record it in data_sources; keep the older frame as history if obtained.
Code scheme	Invented sequence or official IDs	Official IDs plus a permanent internal ID (B2).
Lakes	Inside adjacent units, separate water units, or excluded	Decide with the data owner before loading. Users on lakes are expected (emergency number 110).
Gap and overlap tolerance	Numeric limits for the integrity report	Set after a trial load; start the report in show-only mode, then make it blocking.
Verification	Who confirms a district's data and corrections	Someone other than the person who loaded it.

6. How to build the national geography, step by step

The pipeline below turns official source files into a verified, versioned national dataset. Nothing becomes public until the integrity report is clean and a second person has verified the data.



Units become public only after step 7 (green box). Step 9 (change management) repeats the pipeline for each new vintage.

Step 1

Get the authoritative data in writing

What. Ask NISR (and the land authority if needed) for the village, cell, sector, district and province layers of the chosen vintage, their attribute and code tables, the national outline, and written permission to use them. Record publisher, version, date and licence in data_sources.

Done when. The files are in hand with a written record, and the number of units per level is known.

Step 2

Fix the code scheme

What. Use the official numeric IDs for province, district and sector; extend the same pattern to cells and villages from the official tables. Give every unit a permanent internal ID that is never reused. Add an external_codes table so other systems' codes map in. Derive the display code from these, not by hand.

Done when. A table of unit → code → parent exists for all five levels and the owner has approved it.

Step 3

Stage and clean

What. Load source files unchanged into staging tables. Reproject to WGS84 (EPSG:4326) and keep a metric copy for area and distance. Make geometries valid, convert to multipolygons, and write every fix to a log.

Done when. Staging holds every unit, every geometry is valid, and the log has been read.

Step 4

Build the hierarchy

What. Link each unit to its parent by official codes, not by guessing from geometry. Where codes are missing, match by name inside the parent and confirm that the unit's centre lies inside the parent's polygon. List unmatched units for manual review.

Done when. Every unit has exactly one parent and the unmatched list is empty.

Step 5

Run the integrity report

What. Run the checks in the table below on staging data. Errors block loading; warnings are reviewed.

Done when. The report has no errors and every difference from official counts is explained in writing.

#	Check	Rule	Result needed
1	Counts	Units per level equal the official counts of the chosen vintage. District counts equal published figures (for example Kicukiro 10 sectors, 41 cells, 333 villages; Nyarugenge 10, 47, 350).	Exact match, or each difference explained
2	Parent	Every unit has exactly one parent in the level above.	0 exceptions
3	Inside parent	Part of a child lying outside its parent’s polygon.	At or below agreed tolerance
4	Siblings	Overlap between units of the same level.	At or below tolerance; each overlap listed
5	Coverage	Part of a parent not covered by its children (gaps).	At or below tolerance; each gap explained (lake, border, error)
6	Geometry	Valid, not empty, multipolygon, WGS84, no tiny slivers.	0 exceptions
7	Codes and names	Unique, never reused, correct pattern, names present, spelling normalised.	0 exceptions
8	Outline	Sum of province areas compared with the national outline.	Within agreed tolerance
9	Field spot checks	Known points in every district (offices, landmarks) fall in the expected village.	All pass, or each failure explained

Tolerance values are to be set after the first trial load.

Step 6

Load level by level

What. Order: national outline, provinces, districts, sectors, cells, villages. One transaction per level, the report after each, bulk loading instead of one call per feature. Loaded data is marked pending and is not public.

Done when. All levels are in the database, the report is clean and counts match.

Step 7

Verify

What. Check known landmarks and office locations per district, and random points per district with local staff. Fix, version and re-run. Mark each district verified.

Done when. Every district is verified by someone other than the loader. Only then do public queries see it.

Step 8

Prepare delivery forms

What. Create simplified polygons per level; vector tiles or per-district packs for map and offline use; the names table and search index. Tag each with the dataset version.

Done when. Map and offline packs are generated from the verified version and carry its version number.

Step 9

Change management

What. A new vintage goes through the same pipeline. Produce a difference report (added, removed, renamed, geometry changed), review it, then switch. Keep a mapping from old to new IDs for splits and merges. Keep old versions.

Done when. A trial update (for example the 2012 frame to the 2022 frame) runs end to end.

7. Making the locate engine national

These changes follow from findings A6–A7, B1–B3 and C1–C7.

Area	Change
Answer shape	Return all five levels (province to village) with permanent ID, code and names, read from parent links. Return the confidence per candidate and the unmatched share.
No invented answers	Delete every fixed fallback. A data or database problem returns a real error. A point in a district that is not loaded or not verified returns DATA_NOT_AVAILABLE.
Outside Rwanda	Exact test against the national outline; the rectangle is only a first filter.
Coverage state	Each unit carries: not loaded, loaded (pending), or verified. Public lookups use verified only; the answer says which level it could resolve.
Speed	Bounding-box or distance filter first, exact intersection second; area measured in a metric projection; simplified geometry for the first pass. Benchmark on the full national data and record the result.
Other ways in	Step-by-step choice from real data, place-name search with full path, map tap. Needed when GPS is refused, above 150 m or unavailable.
Offline	Per-district packs of simplified polygons with a spatial index; one cache name used by the service worker and the app; pack version shown to the user.
Tests	A set of known points for each of the 30 districts, plus points near sector, district, lake and national borders, run on every change. Field test on real phones in several provinces; tune thresholds from results.

8. How true data arrives after launch

The system should start with partial but real coverage and grow. It must never fill a gap with sample data. This section describes the mechanisms that let correct data come in once people use it.

8.1 Coverage ladder

Every unit is in one of three states: **not loaded**, **loaded (pending, hidden)**, **verified (public)**. A coverage dashboard shows, per province and district, how many villages are in each state. A person in a not-loaded area sees “this area is not available yet”, and the lookup is counted so that the most requested missing areas are loaded first.

8.2 Citizen correction channel (anonymous)

- A “this is not my village” button lets the person choose the right village from the step-by-step list.
- Store only the reported unit, the claimed unit, the date and an accuracy band. Do not store exact coordinates, which keeps the principle of no location history.
- Aggregate by unit pair. Many reports between the same two villages point to a boundary problem. These appear in an admin review list.
- A “my village is not in the list” report records a possible missing or new village for review.

8.3 Admin verification

- Admins are limited to their own scope (district, sector) and can fix names and spelling. Boundary shape changes go through the pipeline in section 6.
- A second person verifies each change before it becomes public. Every change is recorded (who, what, before, after) in the audit log (E8).

8.4 Layers that arrive later

These are separate from the boundary backbone and attach to a unit by its permanent ID: people and their contacts, local offices (the 2022 census office locations can seed this), local emergency numbers, photos. Each record has a source and a verifier. If boundaries change, the old-to-new ID mapping moves them. None of these is needed to make the geography work.

8.5 New boundary versions

When NISR or the land authority publish a new set, it enters through step 9: difference report, review, switch, with history kept.

8.6 What to measure after launch

- Lookup success rate by district; share of low-confidence, gap and not-available answers.
- Corrections per 1,000 lookups, by unit pair.
- Response time (typical and slow cases) on national data.
- Share of villages loaded and verified, by district.

9. Roadmap and acceptance criteria

Phase	Work	Done when
0. Stop the fakes, close access	Remove the fixed fallback and mock data from public routes. Remove reveal and dossier from public routes. Real authentication. Revoke public grants on leaders. Make setup scripts match documents.	With an empty database every public route returns an honest “not available” and never sample data. A call without login cannot write or read private fields (tested).
1. Foundation for national data	Code scheme; schema with permanent IDs, versions, sources, verification, names, parent links and national outline; rewritten ingest; integrity report; numbered migrations.	A trial load of one province passes the report. The schema can be re-run.
2. National boundary load	Section 6, steps 1–7, for all provinces.	Counts match, report clean, every district verified by a second person, licence recorded.
3. National locate engine	Section 7: parent-link hierarchy, exact outline test, coverage states, speed work, other ways in, known-point tests.	Known points for all 30 districts pass. Response-time target measured and met on national data. DATA_NOT_AVAILABLE works for unloaded areas.
4. Admin, verification, corrections	Scoped admin with audit; verification queue; citizen correction channel; coverage dashboard.	An admin verifies a unit and it becomes public. Corrections reach the queue. The audit log shows every change.
5. Map, search, offline, language	Simplified geometry or tiles; name search; district offline packs; Kinyarwanda and English.	A downloaded district works with the network off. Search separates duplicate names.
6. Operations	Tests and CI, health check, rate limits, monitoring, backups with tested restore, rollback, load test, real-phone tests in several provinces.	Each item is demonstrated with evidence.

> Order matters. Phases 0 and 1 come first because every later phase depends on honest behaviour and on a correct code scheme. People and contact data (section 8.4) can be loaded at any point after phase 1 and does not block the geography.

10. What I could not determine

- **The state of the live site and the GitHub repository.** The specification names a repository, a branch and a live address. The archive does not match the specification’s file list, so I do not know which version is deployed.
- **Whether the code runs.** I read it; I did not install or run it. Behaviour statements come from reading.
- **Licence and download availability** of the boundary layers, and their attribute tables (section 5.2).
- **Which frame is current** (2,148 / 14,837 or 2,146 / 14,774).
- **Cell and village counts** for most districts; only Kicukiro and Nyarugenge were checked against public pages.
- **Real phone behaviour** of the accuracy values and of the call and WhatsApp links.

11. Public sources consulted (29 October 2026)

Pages were read for facts and counts. They should be re-checked before being relied on.

- National Institute of Statistics of Rwanda: district baseline publications (2,148 cells, 14,837 villages) — statistics.gov.rw
- Rwanda Space Agency GIS portal, Administration_Rwanda layers (sector and cell descriptions, census mapping) — gis.space.gov.rw
- REMA GIS server, Administrative_Boundaries service (District, Sector, Cell, Village; RLMUA/RNRA 2012) — gis.rema.gov.rw
- RBC GIS, Full_District_boundary_2022 (boundaries from 2022 census village boundaries) — gis.rbc.gov.rw
- FAO, Rwanda Population and Housing Census 2022, agricultural module metadata (2,146 cells, 14,774 villages) — openknowledge.fao.org
- Rwanda Space Agency GIS, Government_Assets (office locations from 2022 census mapping) — gh.space.gov.rw
- NISR sector boundaries service description (official ID scheme) — infrastructure.space.gov.rw
- Nyarugenge District website (10 sectors, 47 cells, 350 villages) — nyarugenge.gov.rw
- Kicukiro District, public encyclopedia entry (10 sectors, 41 cells, 333 villages)
- Ministry of Health data reference page listing cell and village shapefiles — docs.nhic.moh.gov.rw
- NISR Comprehensive Food Security and Vulnerability Analysis village questionnaires, 2015 and 2018 (households per village) — microdata.statistics.gov.rw
- Open-source Rwanda geography package documentation (duplicate place names) — rubydoc.info/gems/rwanda